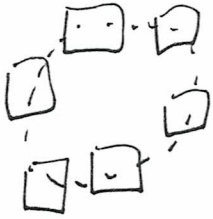


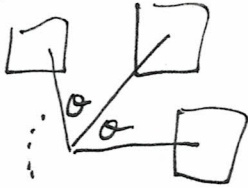
Need to arrange N items in a circle, maximizing their size but with no overlap.

① space items evenly around a circle.

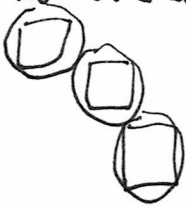


② Angle between items

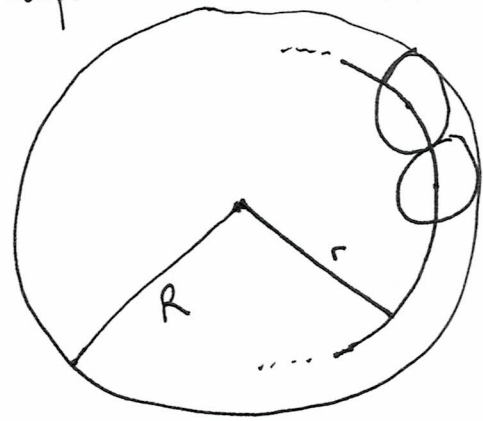
$$\theta = \frac{2\pi}{N}$$



③ Circular buffer zone around each item ensures no overlap.



④ Compute r based on total width



$$R = \frac{D}{2}$$

$$R = r + r'$$

$$R = r + r \cdot \sin \frac{\theta}{2} = r (1 + \sin \frac{\theta}{2})$$

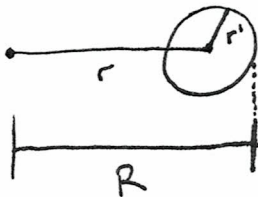
$$r = \frac{R}{1 + \sin \frac{\theta}{2}} = \frac{D/2}{1 + \sin \frac{\theta}{2}}$$

⑤ Setup.

r = radius of main circle

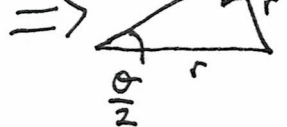
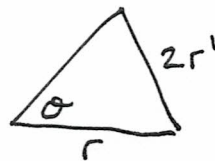
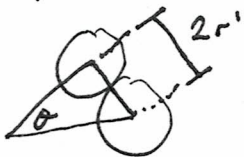
r' = radius of buffers

R = radius of whole layout



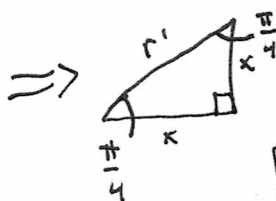
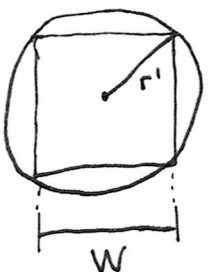
$$R = r + r'$$

⑥ Compute r'



$$r' = r \cdot \sin \frac{\theta}{2}$$

⑦ compute size of inscribed square.



$$x = \frac{W}{2}$$

$$x = r' \cdot \sin \frac{\pi}{4}$$

$$W = \frac{2r'}{\sqrt{2}}$$